

Venkata Naga Sai Vyshnavi Kurivella

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SUMMARY

Software Developer with 3 years of experience in Python and SQL, specializing in scalable web applications and data-driven backend systems. Proficient in Django, Flask, NumPy, and Pandas, with strong expertise in MySQL, SQL Server, and SQLite. Skilled in deploying Python applications on AWS and Azure, and experienced in Agile and SDLC practices. Adept at using Git/GitHub for version control and building responsive UIs with HTML, CSS, JavaScript, and Bootstrap.

PROFESSIONAL EXPERIENCE

PLATFORM DEVELOPER / SOFTWARE DEVELOPER

Jan 2025 - Current

Uber, IL

- Led backend development of Uber's internal expense reconciliation platform using Python and Flask, delivering scalable RESTful APIs to process third-party vendor data.
- Built data pipelines with NumPy, Pandas, and Excel macros to automate financial data cleaning and transformation, reducing manual effort by 70%.
- Designed and optimized MySQL schemas and queries, improving API response times by 40% for high-volume financial transactions.
- Deployed applications on AWS EC2, RDS, and integrated Lambda functions for serverless execution, enabling a scalable and cost-effective infrastructure.
- Containerized services using Docker and managed deployments with Kubernetes, ensuring resilient and efficient delivery across environments.
- Utilized PyCharm, VS Code, and Postman to streamline backend API testing, debugging, and efficient code navigation throughout the development lifecycle.

ASSOCIATE SOFTWARE ENGINEER / SOFTWARE DEVELOPER

Jun 2021 – Nov 2022

DXC Technologies India Pvt. Ltd, India

- Developed and maintained web-based insurance illustration platforms using Python and Django, automating over 500 personalized policy illustrations with dynamic rider and premium integration.
- Designed scalable backend logic using Python OOP and Django REST Framework (DRF), enabling complex insurance computations and standardized API responses.
- Built and integrated RESTful APIs for real-time data exchange with external insurance services, improving system flexibility and processing speed.
- Ensured regulatory compliance by automating validation rules in collaboration with actuaries, eliminating compliance gaps in all illustration outputs.
- Optimized SQL Server schemas and stored procedures, reducing query time and enhancing performance across data reporting and illustration modules.
- Leveraged Pandas and NumPy for data transformation pipelines supporting fund analytics, pricing models, and cross-policy performance comparisons.
- Provisioned AWS EC2 and S3 for secure deployment and data storage, and managed CI/CD pipelines via Jenkins for smooth releases across platforms.
- Collaborated in Agile teams, actively participating in sprint planning and code reviews while maintaining Git-based version control workflows.
- Employed Jupyter Notebook for prototyping and validating data transformation scripts using Pandas and NumPy in early-stage pipeline development.

PYTHON DEVELOPER

Aug 2020 – May 2021

KPMG, India

- Engineered backend components for a logistics and inventory system using Python, Django, and Django, enabling seamless integration with MongoDB for managing unstructured vendor and product data.
- Created RESTful APIs to support real-time interactions between client-side applications and backend logic, enhancing inventory tracking and operational efficiency.
- Applied Pandas and NumPy to transform and analyze high-volume transactional datasets, delivering insights that informed supply chain improvements.

- Restructured SQL Server and MongoDB data models to boost query performance and support scalable hybrid database operations.
- Assembled dynamic UI elements with HTML, CSS, Bootstrap, and JavaScript, and enriched interactivity through AJAX for responsive user experiences.
- Engaged in Agile workflows, including sprint planning and retrospectives, to align development cycles with evolving business needs.
- Worked in cross-platform environments (Windows and Linux) to deploy, test, and manage backend services and automation scripts.

EDUCATION

MASTER OF SCIENCE IN COMPUTER SCIENCE <i>University of South Florida</i>	Jan 2023 - Dec 2024 Tampa, US
BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING <i>KKR and KSR Institute of Technology and Sciences</i>	Jul 2017 - Jun 2021 Guntur, India

TECHNICAL SKILLS

Programming Languages:	Python, SQL, JavaScript
Web Frameworks:	Django, Flask, Django REST Framework(DRF)
Data Libraries:	Pandas, NumPy, Excel Macros
Databases:	MySQL, SQL Server, SQLite, MongoDB (via Djongo)
API & Data Protocols:	RESTful APIs, JSON, XML, AJAX
Web Technologies:	HTML5, CSS3, Bootstrap, jQuery
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda), Azure, Docker, Kubernetes
CI/CD & Version Control:	Jenkins, GitHub, GitHub Actions
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook, Postman
Methodologies:	Agile (Scrum), SDLC, Waterfall
Operating Systems:	Windows, Linux(Ubuntu/CentOS)

PROJECTS

<i>DocInsight – Empowering Informed Healthcare Choices / React</i> Built and integrated search and recommendation features for a React-based healthcare platform. Improved user engagement and doctor-patient match accuracy through real-time data, AI-driven suggestions, and enhanced UI/UX components. <i>Machine Learning Model Optimization / Python</i> Optimized CNNs for CIFAR-100 image classification, achieving over 44% test accuracy. Enhanced a baseline model through 15-epoch training, class-wise performance analysis, and implemented a snapshot ensemble of 5 models to boost overall accuracy in a complex multi-class environment. <i>Vendor Reconciliation API / Flask, MySQL, AWS Lambda</i> Built a microservice to automate invoice reconciliation for a finance department, reducing manual efforts by 70%. Utilized Flask for API logic, AWS Lambda for serverless processing, and deployed on AWS EC2 with a MySQL backend. <i>Financial Forecasting System / Django, Pandas, PostgreSQL</i> Engineered backend services for a financial forecasting tool using Django, Pandas, and PostgreSQL. Implemented time-series analysis models and developed RESTful APIs to serve historical and projected financial data for internal business intelligence.
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